A Comparative Analysis of ML Models for Breast Cancer Diagnosis

STSCI 6520 Capstone Project Proposal

K. Chan

October 30, 2025

1. Motivation

The early and accurate diagnosis of breast cancer is a critical task in medical treatment. Using Machine Learning (ML) methods can significantly increase accuracy alongside clinical judgement. The Breast Cancer Wisconsin (Diagnostic) Dataset provides a high-dimensional snapshot of tumor characteristics which is applicable to multiple ML methods, which can be compared to show which method gives the best potential to diagnose cancer as benign or malignant.

While predictive accuracy is the goal for ML methods it is not the only goal in medical treatment. Doctors and especially patients may not even use the decision of a machine model if they don’t understand how it makes a prediction. This is the reason for this proposed study; comparing "black box" models that have high accuracy but go beyond most intuitive understanding to simpler but powerful models that can be interpreted, but may not leverage all the complexities and interactions provided by data.

This project directly addresses this trade-off. I will explore how evaluation metrics beyond simple "accuracy" can bridge this gap. This also includes understanding how ML and Medical terms like **false negative** (a malignant tumor classified as benign) might be considered differently for an ML scientist and a doctor; leading to difference decisions. Are metrics like **Precision** (of all patients the model flagged as "malignant," how many actually were?) and **Recall** (of all true malignant cases, how many did the model find?) better at giving intuitive understanding for people making medical decisions. This project seeks to find the model that provides the best *balance* between predictive power and interpretability for this critical problem.

2. Research Problem

This project will use the **Breast Cancer Wisconsin (Diagnostic) Dataset** (BCWDD) to cross-compare three classification models. ML classification is frequently used but often not well understood especially by laypersons due to algorithmic complexity and black box methods. Comparing models on the same dataset provides more intuitive understanding of key features in model implementation and evaluation, like validation and verification, parameter tuning, and what metrics are best for understanding and communication. ML methods are becoming more prevalent and more accessible via advanced AI tools; it’s worthwhile to learn to compare and question the criteria for a model - even for those who won’t build them. Cancer diagnoses and decisions around critical medical care like surgery or chemotherapy are common to many people. Is language like '"low probability of false positives" really interpretable to people caring for loved ones? What other ways of communicating could be considered?

The models under selection are fully detailed in the next section, briefly, they are; **Random Forest, for accuracy and robustness, Lasso Regression** selected for interpretability, and **Support Vector Machines**, as a non-linear comparison to Regression. The models performance will be compared on accuracy and interpretability on a meaningful medical dataset.

3. Proposed Steps and Expected Outcomes

All analysis will be conducted using the **R** programming language. The BCWDD data contains the target variable; **diagnosis**, and 30 numerical predictors derived from tumor imaging data, (radius, texture, density etc…). The predictors are numeric and will require preprocessing steps like numeric standardization and mean centering. Best practice ML methods will be used for each - including k-fold cross validation (k=10; 10 groups to fit models). The predictions from each classification model can generate a Confusion Matrix, the number of true false, misses and matches against the diagnoses. These numbers then create the metrics that can be compared for model performance; Accuracy, Precision, Recall and likely the most interpretable - F1-Score.

As an expected outcome, the "black box models" Random Forest and SVM should have the highest performance, however, the potential for complex and non-linear relationships is very high with 30 numeric predictors. This means the models will lack **interpretability** - the ability to be understood. In this sense Lasso Regression will likely be the most interpretable. What remains to be seen is if the potential gap in accuracy is worth providing better understanding for people making decisions.

References

[1] Al-Dhief, F. T., et al. (2024). *A Machine Learning Approach for Breast Cancer Prediction using Random Forest Algorithm*.

[2] Asri, H., et al. (2016). *SVM and SVM Ensembles in Breast Cancer Prediction*.

[3] Wu, J., et al. (2018). *Logistic LASSO regression for the diagnosis of breast cancer using clinical demographic data and the BI-RADS lexicon for ultrasonography*.